A Coreset Selection Framework with Ensemble Aggregation for Image Classification: Supplemental Document

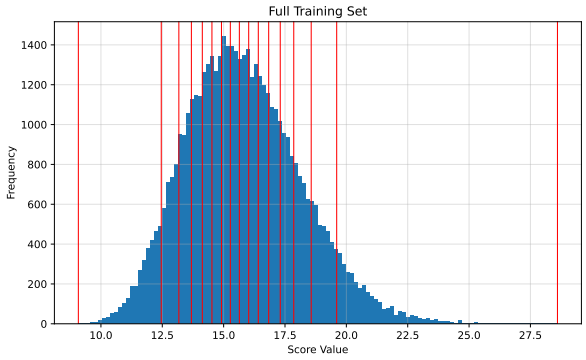
1. CODE AND IMPLEMENTATION

The code of our proposed approach is available here: `https://github.com/PedroRochaDantas/SCOSS`.

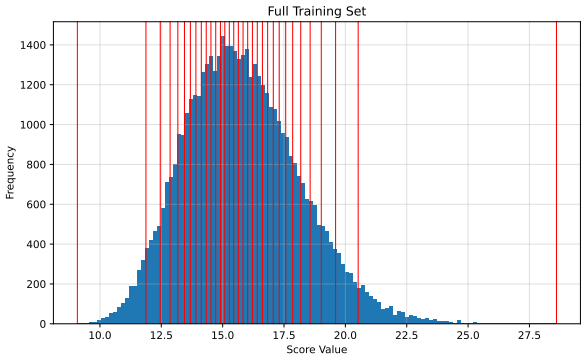
2. ADDITIONAL RESULTS AND VISUALIZATIONS

A. Score Interval Partitioning

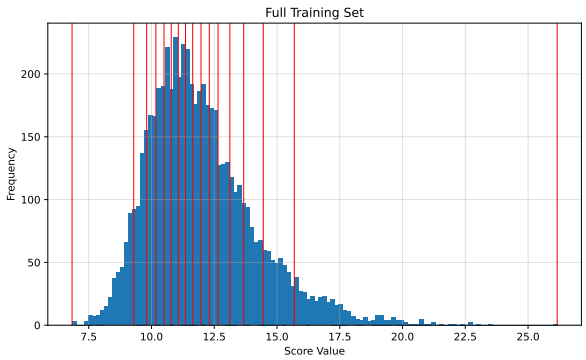
Figure S1 illustrate the score interval partitioning adopted by the proposed method for the CIFAR-10 and CUB-200 datasets using 16 and 32 intervals. The vertical lines indicate the interval boundaries computed from the score distribution of the full training set. These visualizations demonstrate how the adaptive partitioning divides the score space into regions containing approximately the same number of samples, allowing the subsequent sampling procedure to preserve the original score distribution while ensuring that all score regions are represented.



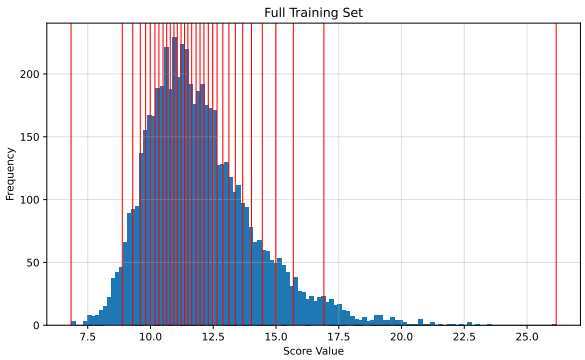
(a) CIFAR-10 with 16 intervals.



(b) CIFAR-10 with 32 intervals.



(c) CUB-200 with 16 intervals.



(d) CUB-200 with 32 intervals.

Figure S1: Score interval partitioning for the CIFAR-10 and CUB-200 datasets using 16 and 32 intervals.

B. Sampling Score Histograms

To provide additional insight into the sampling strategies, Figures S2 and S3 show the histograms of the sampling scores obtained for the CUB-200 and CIFAR-10 datasets, respectively. The figures compare the score distribution of the complete training set with those produced by each sampling strategy. These visualizations illustrate how each method selects samples across the score range, highlighting whether the selected subset preserves the overall distribution or emphasizes specific score intervals.

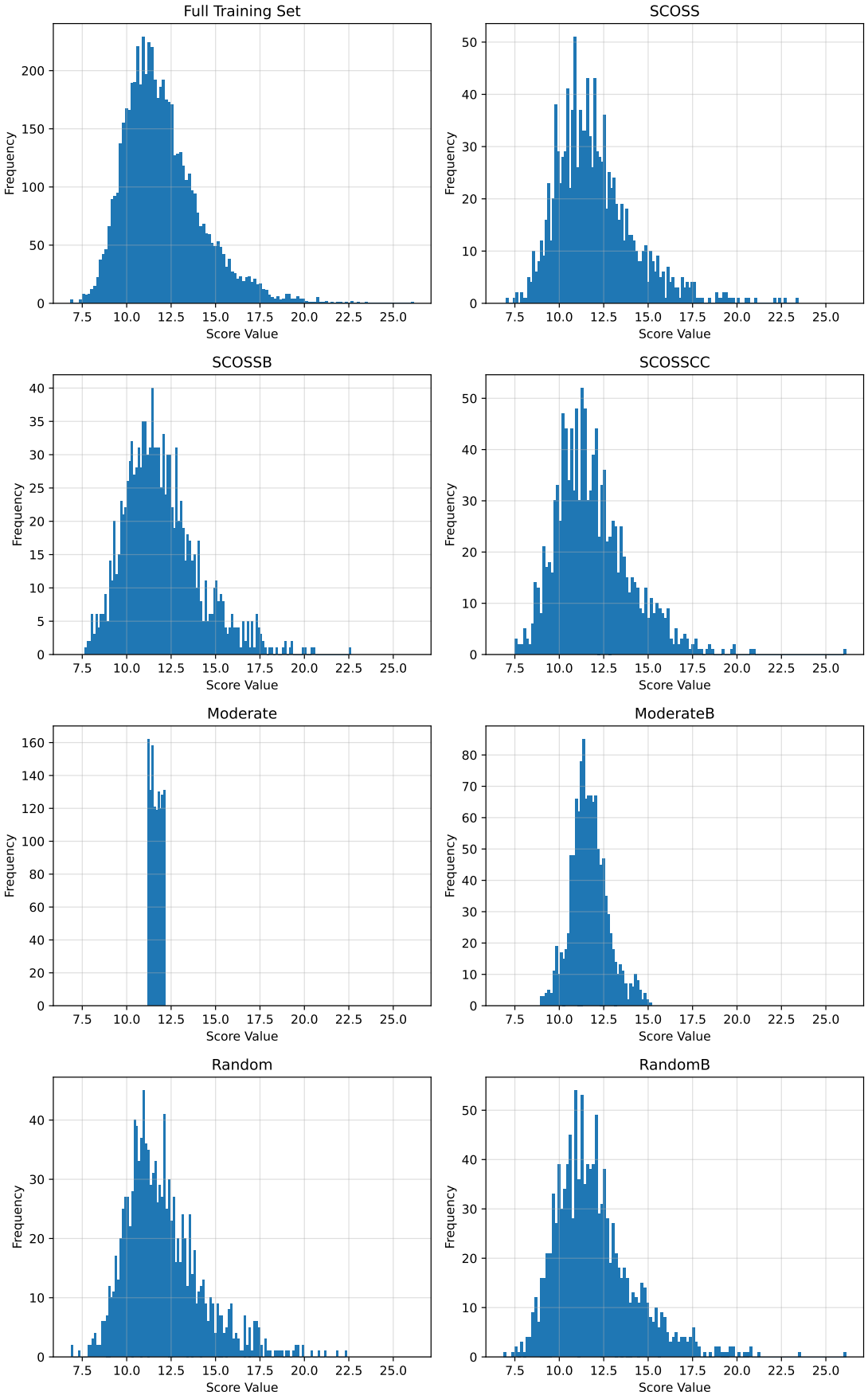


Figure S2: Sampling score distribution for the CUB-200 dataset. (20% ratio)

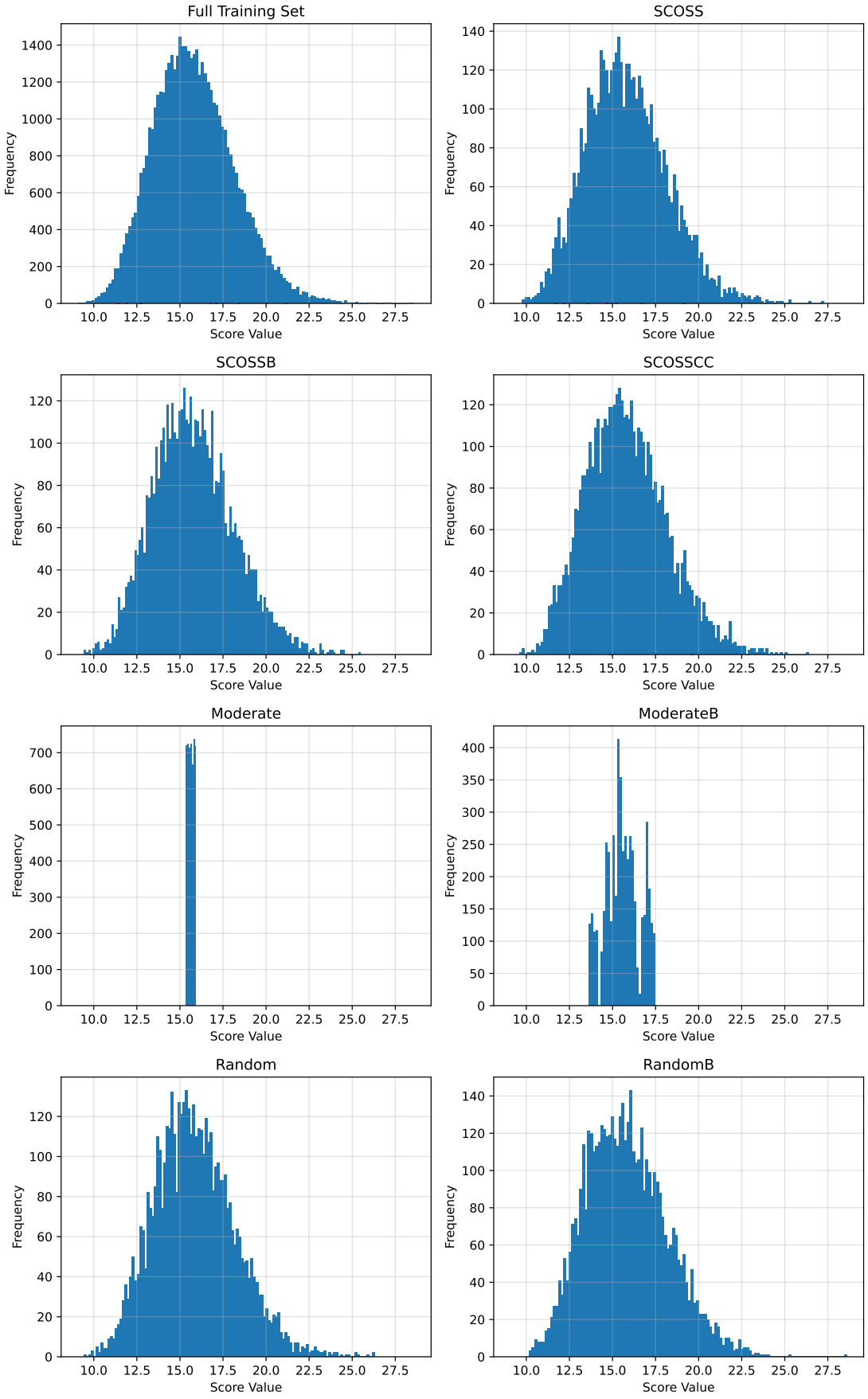


Figure S3: Sampling score distribution for the CIFAR-10 dataset. (10% ratio)